$U = A \oplus B + C \cdot B$

XOR OR

A	B	C	U
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$A \oplus B$ Vero

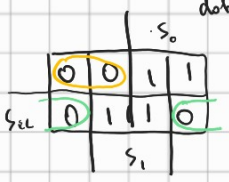
→ uso la mappa di Karnaugh



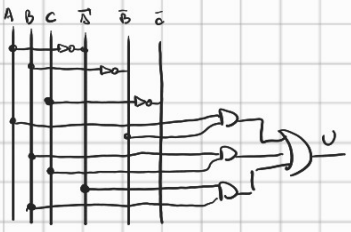
sintesi minima AND/OR

$U = A\bar{B} + BC + \bar{A}B$

se voglio fare la mappa di Karnaugh con SM? dobbiamo farne il duale

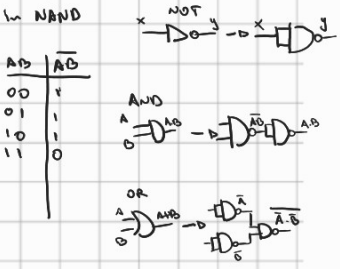






{NOT, AND, OR}

voglio fare l'implementazione solo in NAND



A B	$\overline{AB}$
0 0	1
0 1	1
1 0	1
1 1	0

$\overline{\bar{A} \cdot \bar{B}} = A + B$

$U = (SEL + S_0) \cdot (\overline{SEL} + S_1)$

si può usare anche l'Adiacenza Logica

Ma perché dobbiamo farlo?

viene fuori da

$\underbrace{(SEL + S_0 + S_1)}_{T} \cdot \underbrace{(S_{el} + S_0 + \bar{S}_1)}_{T} = (S_{el} + S_0)$

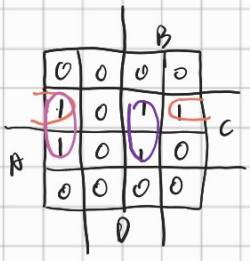
grazie al processo di induzione e riusciamo a semplificare il tutto

$(T + S_1) \cdot (T + \bar{S}_1)$

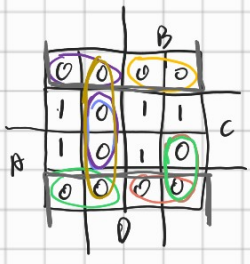
$T \cdot T + T \cdot \bar{S}_1 + S_1 \cdot T + S_1 \bar{S}_1 =$

$= T + T(S_1 + S_1) + 0$

$= T + T \cdot 1 = T + T = T$



AND-OR



OR-AND

$U = C \cdot (\bar{A} + \bar{B} + D) \cdot (B + \bar{D})$

$U = BCD + \bar{A}C\bar{D} + \bar{B}C\bar{D}$

Voglio costruire una porta logica programmabile

4 ingressi

A, B, S_0, S_1

IF S_0 S_1	U
0 0	A · B
0 1	A + B
1 0	$\bar{A}$
1 1	-

indifferenza → ci metto quello che mi viene più comodo

S_0 S_1	S_0	S_1	A	B	U
(0,0)	0	0	0	0	0
	0	0	0	1	0
	0	0	1	0	0
	0	0	1	1	0
(0,1)	0	1	0	0	0
	0	1	0	1	1
	0	1	1	0	1
	0	1	1	1	1
(1,0)	1	0	0	0	1
	1	0	0	1	1
	1	0	1	0	1
	1	0	1	1	1
(1,1)	1	1	0	0	-
	1	1	0	1	-
	1	1	1	0	-
	1	1	1	1	-

indifferenze sulle USCITE

S_0 S_1	S_0	S_1	A	B	U
(0,0)	0	0	0	0	0
	0	0	0	1	0
	0	0	1	0	0
	0	0	1	1	0
(0,1)	0	1	0	0	0
	0	1	0	1	1
	0	1	1	0	1
	0	1	1	1	1
(1,0)	1	0	0	0	1
	1	0	0	1	1
	1	0	1	0	1
	1	0	1	1	1
(1,1)	1	1	0	0	-
	1	1	0	1	-
	1	1	1	0	-
	1	1	1	1	-

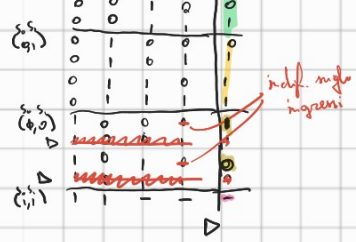
indiff. negli ingressi

posso anche togliere B

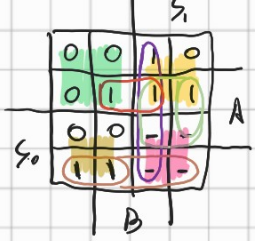
esistono anche INDIFFERENZE SULLE USCITE

S_0 S_1	S_0	S_1	A	B	U
(0,0)	0	0	0	0	0
	0	0	0	1	0
	0	0	1	0	0
	0	0	1	1	0
(0,1)	0	1	0	0	0
	0	1	0	1	1
	0	1	1	0	1
	0	1	1	1	1
(1,0)	1	0	0	0	1
	1	0	0	1	1
	1	0	1	0	1
	1	0	1	1	1
(1,1)	1	1	0	0	-
	1	1	0	1	-
	1	1	1	0	-
	1	1	1	1	-

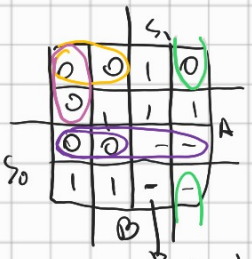
AND-OR



→ Mappa di K.



$U = AB\bar{S}_0 + S_1A + S_1B + S_0\bar{A}$

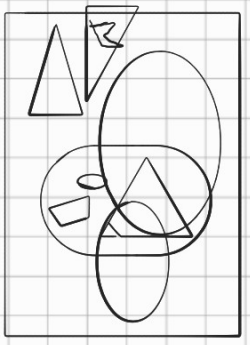


OR - AND

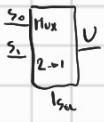
$U = (\bar{S}_0 + \bar{A}) \cdot (S_0 + S_1 + B) \cdot (S_0 + S_1 + A) \cdot (A + \bar{S}_1 + B)$

$U = (\bar{S}_0 + \bar{A}) \cdot (S_0 + S_1 + B) \cdot (S_0 + S_1 + A) \cdot (A + \bar{S}_1 + B)$

questo diventa 0 perché NON CI TORNA UTILE

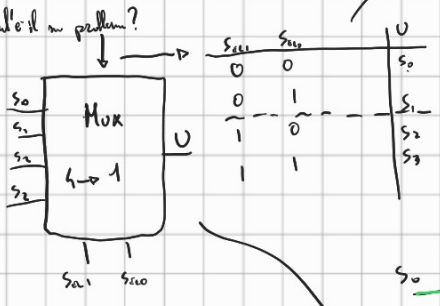


Multiplexer, DeMUX, Decoder



Qual'è il mio problema?

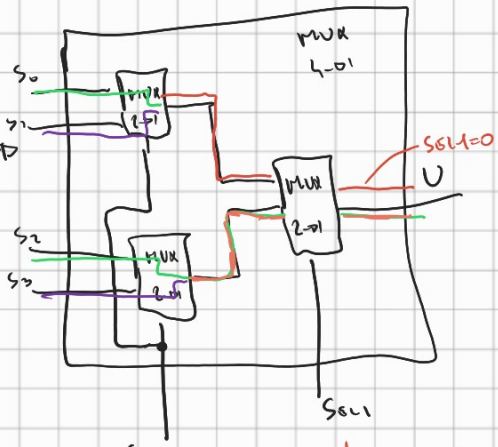
Tabella di verità incomplete



MUX n° selettori

2 → 1

ingressi

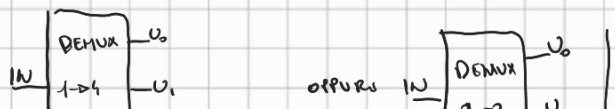


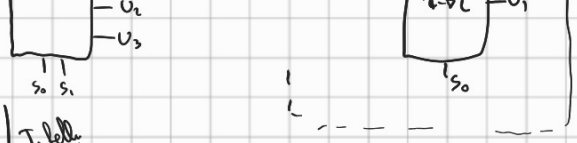
Sel0=0
Sel1=1

Sel1=0

Sel1=1

DeMUX:

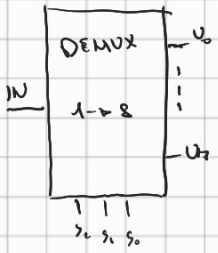




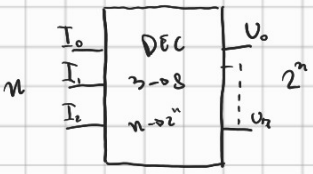


S_1	S_0	IN	U_0	U_1	U_2	U_3
0	0	0	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	1	0	1	0	0
1	0	0	0	0	0	0
1	0	1	0	0	1	0
1	1	0	0	0	0	0
1	1	1	0	0	0	1

S_1	S_0	U_0	U_1	U_2	U_3
0	0	IN	0	0	0
0	1	0	IN	0	0
1	0	0	0	IN	0
1	1	0	0	0	IN



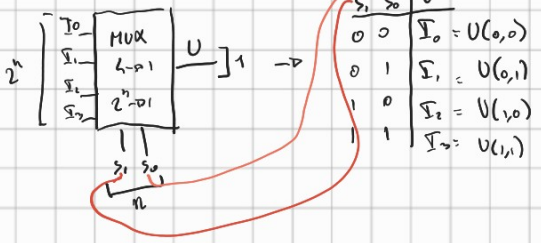
Decoder o Dec



I_2	I_1	I_0	U_0	U_1	...	U_7
0	0	0	1	0		0
0	0	1	0	1		0
0	1	0	0	0		0
0	1	1	0	0		0
1	0	0	0	0		0
1	0	1	0	0		0
1	1	0	0	0		0
1	1	1	0	0		1

Perché il fatto vedere queste componenti?

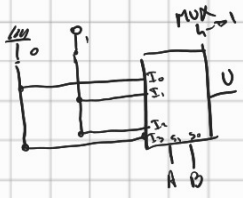
→ la funzionalità del MUX 4->1



S_1	S_0	U
0	0	$I_0 = U(0,0)$
0	1	$I_1 = U(0,1)$
1	0	$I_2 = U(1,0)$
1	1	$I_3 = U(1,1)$

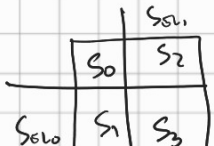
può fare qualsiasi funzione di n variabili

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0



MUX → porta logica UNIVERSALE

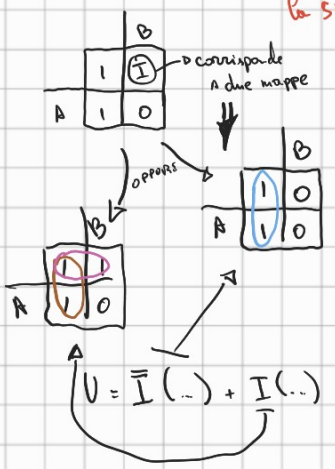
S_{el1}	S_{el0}	U ← uscita
0	0	S_0 ← ingresso
0	1	S_1
1	0	S_2
1	1	S_3



Mappe a variabili riportate

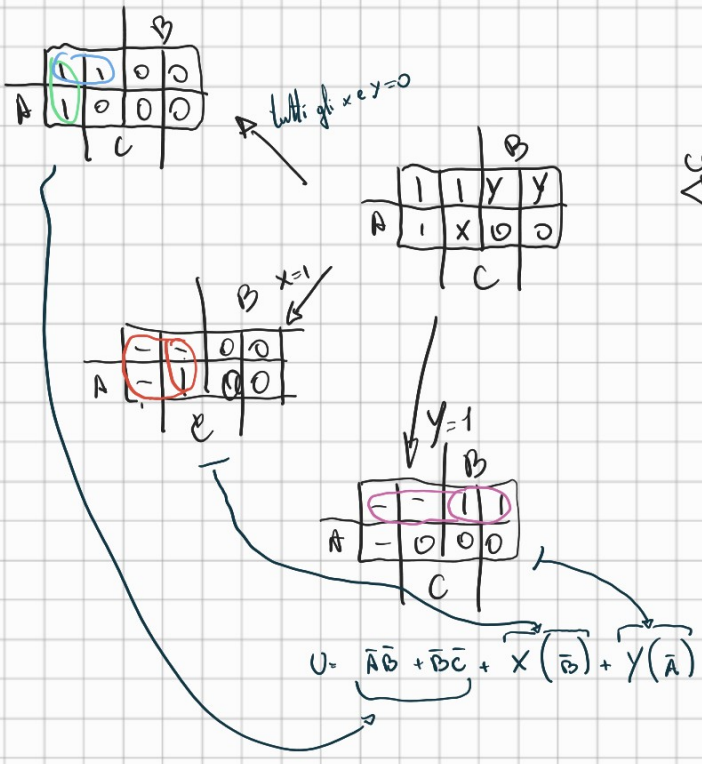
Ingressi possono comparire anche dentro delle celle delle mappe di K

Come individuiamo le soluzioni? PERÒ ▷ NON OTTENIAMO LA SINTESI MINIMA ma la SINTESI QUASI MINIMA

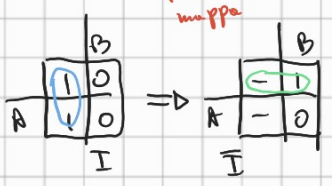


$= \bar{I}(\bar{B}) + I(\bar{B}+\bar{A})$ (OR)

Però NON HA SENSO perché prendo due fole che abbiamo già preso in un'altra mappa



complichiamo



$U = \bar{B} + \bar{I}(\bar{A})$

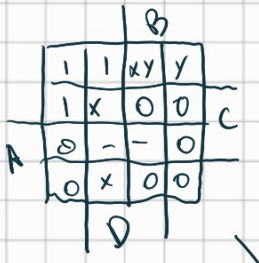
non ci sono I da moltiplicare pk sono già inclusi nella MAPPA

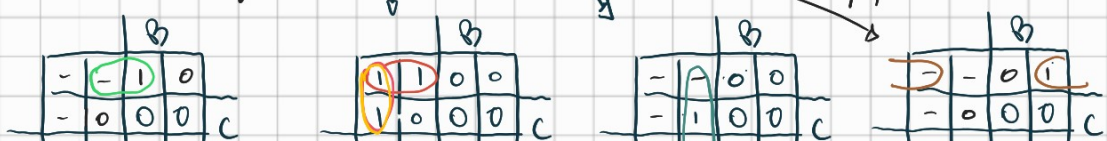
Le regole per fare questo esercizio:

1) Si individuano le variabili riportate
2) Si fa la mappa con tutte le var. rip. = 0 e le mappe con 1 var. rip. per volta = 1
3) Si sommano (OR) la mappa con tutte (variabili riportate) + V.R. che moltiplica la (AND) sua mappa.

variabili riportate sono 3 {x, y, xy}

(0→0, -→-, 1→-) ← differenze

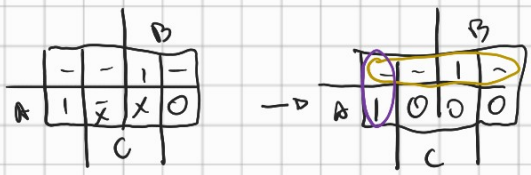




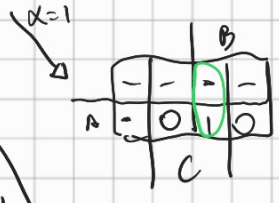


$xy(D\bar{A}\bar{C})$ + $\bar{A}\bar{B}\bar{D} + \bar{A}\bar{C}\bar{B}$ + $x(\bar{B}D)$ + $y(\bar{C}\bar{A}\bar{D})$

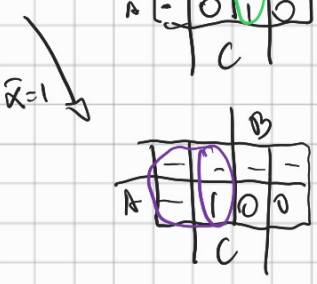
$$\bar{A}\bar{B}\bar{D} + \bar{A}\bar{C}\bar{B} + x(\bar{B}D) + y(\bar{C}\bar{A}\bar{D}) + xy(D\bar{A}\bar{C})$$



$\bar{A} + \bar{C}\bar{B}$



$x(CB) \rightarrow \bar{A} + \bar{C}\bar{B} + x(CB) + \bar{x}(B)$



$\bar{x}(\bar{B})$

Multiplexer

